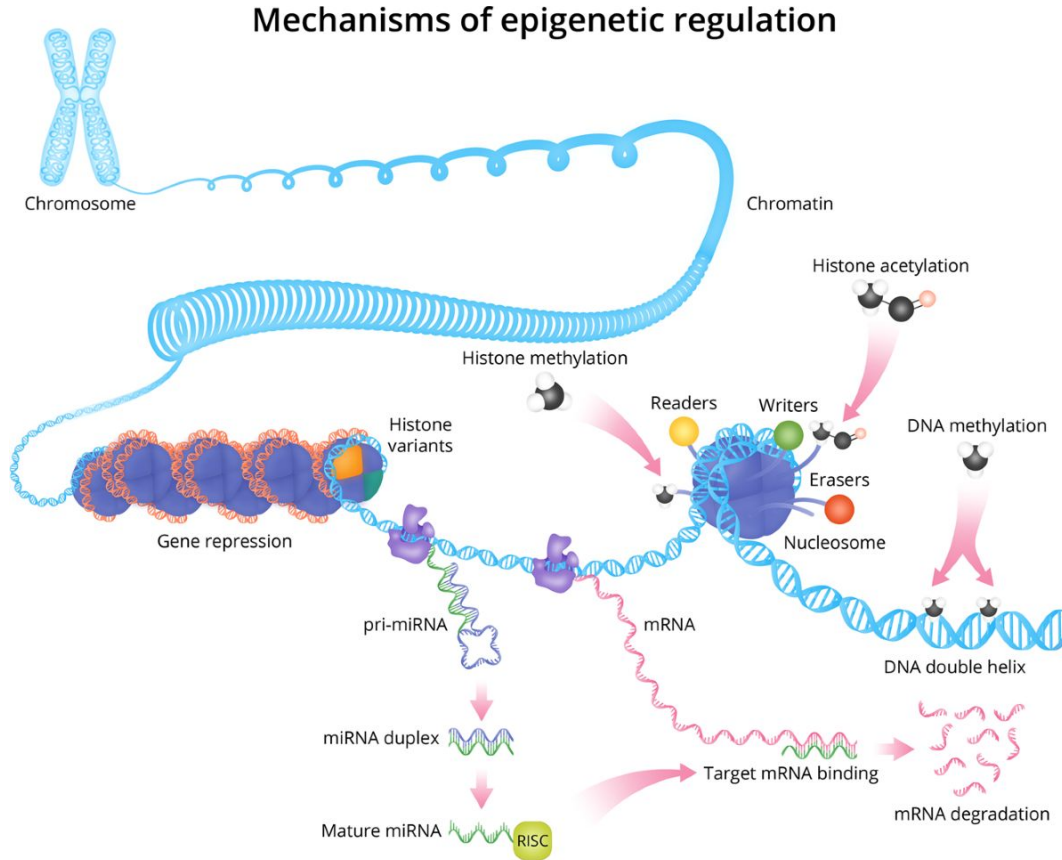


Epigenomics Data Analysis: DNA Methylation

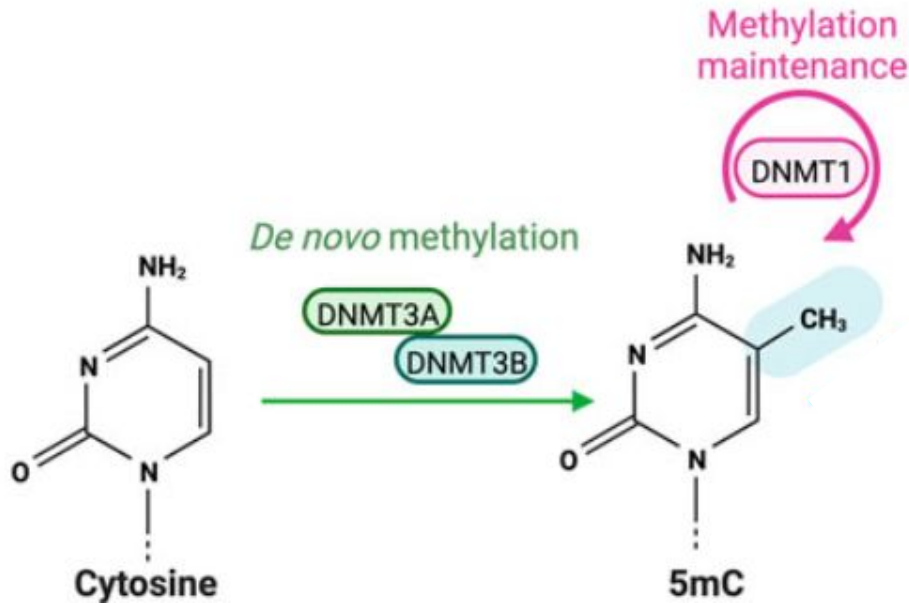
Louella Vasquez

26-10-05

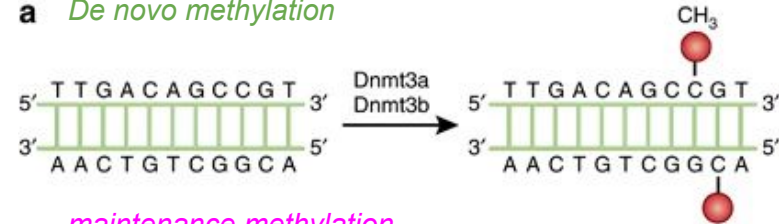
Epigenome regulates gene expression via chromosomal alteration that does not involve changes in the DNA sequence



DNA methylation is a stable, heritable chemical modification

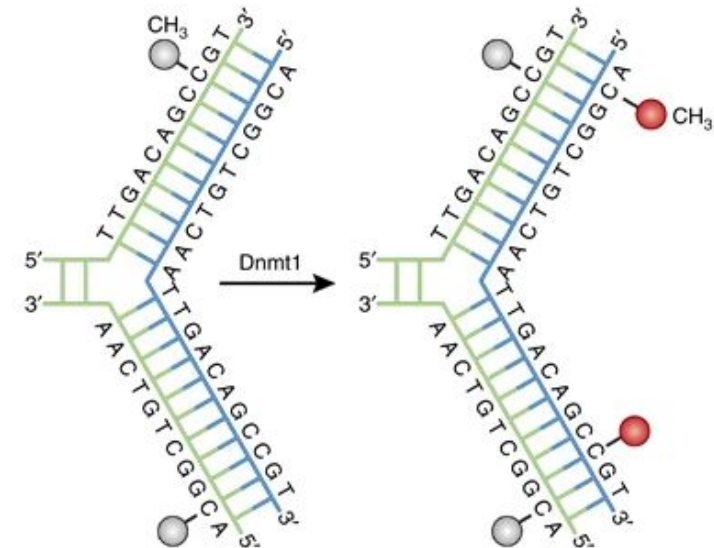


a *De novo methylation*



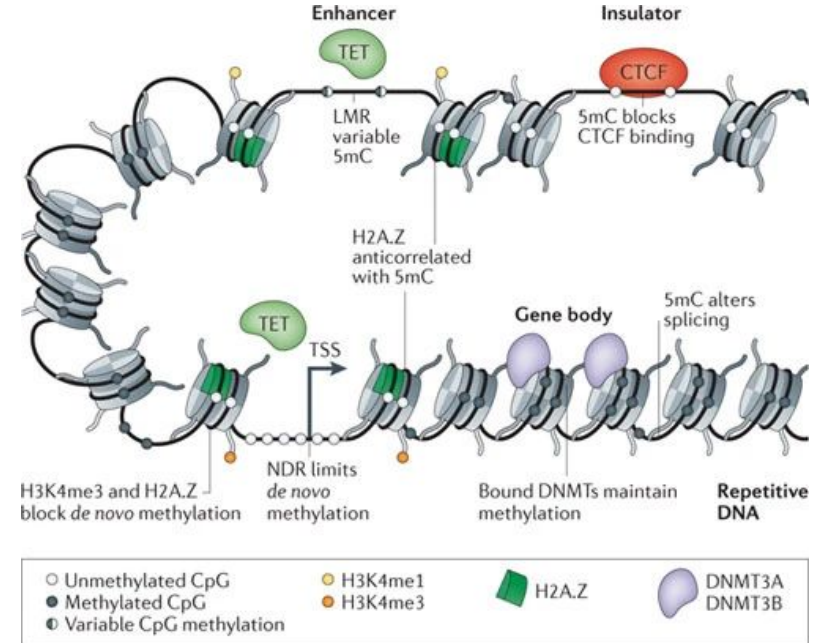
b

maintenance methylation



5mC in the genome

- predominantly in CpG (Cytosine-phosphate-Guanine) dinucleotides in metazoan genomes
 - ~28M CpGs in humans
 - 60–80% methylated in somatic cells
- CpGs in CG-dense regions are CpG islands (CGIs)
 - 200-2000 bp with >50% GC-content
 - CGIs tend to be in promoters, unmethylated for transcribed genes
- non-CpG methylation has been mainly observed in hESCs and neuronal cells in humans
 - CHH, CHG where H = A, C or T



Nature Reviews | Genetics

Key functions of DNA methylation

Tissue specific gene regulation

Suppression of transposable elements

Essential for normal development

Genomic imprinting

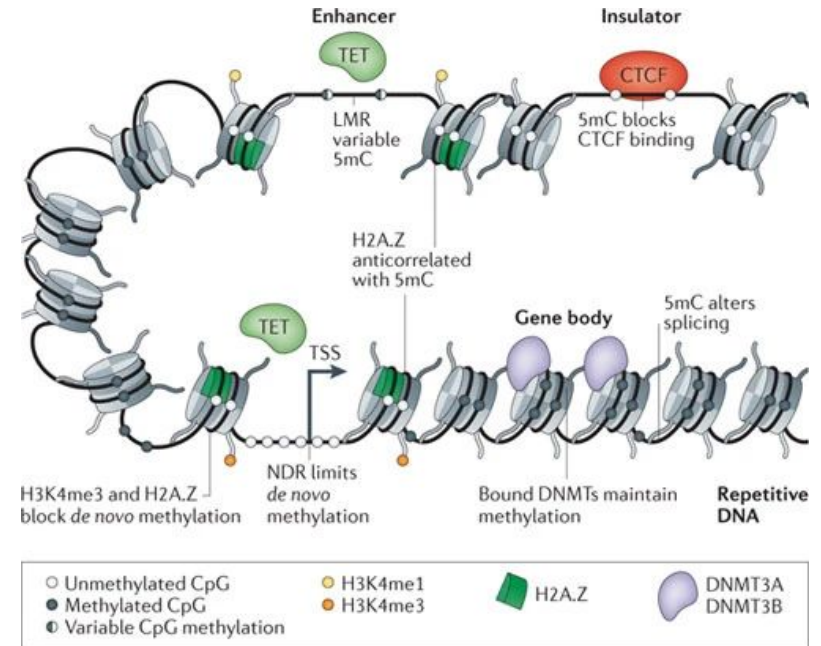
X-chromosome inactivation

Ageing

- Global hypomethylation is proportional to age

Cancer

- Global hypomethylation and locus-specific hypermethylation of CpG islands



Nature Reviews | Genetics

DNAM measurement

Differentiate mC from C

C > T Bisulfite conversion

A: 5'-GACC^{CT}CCAGGTC^{CA}GC^{CT}-3'

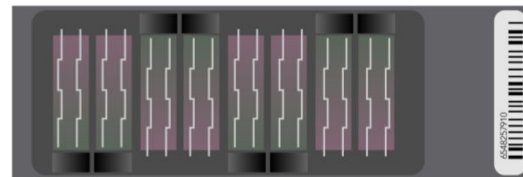
B: 3'-CTGG^{CA}AGGTC^{CA}GGTC^{CA}CGA-5'

A: 5'-GATC^GTTTTAGGTTAGTAGT^GGTT-3'

B: 3'-TTGG^{CA}AGGTTAGGTTGTTAT^GCA-5'



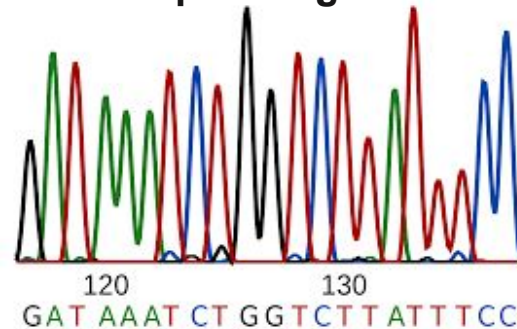
DNA amplification



Methylation Array

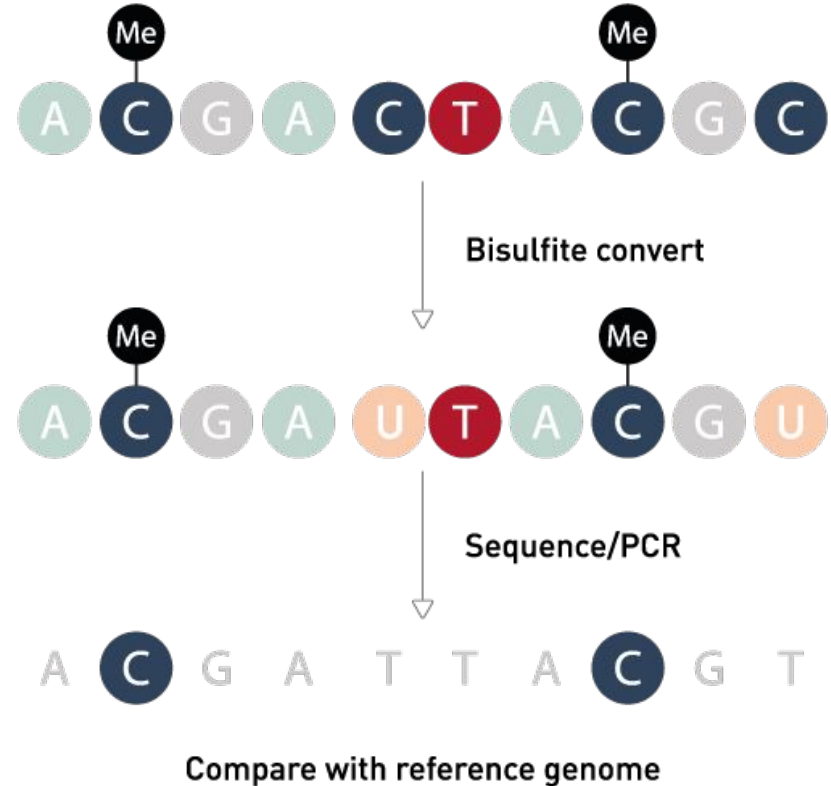


DNA Sequencing

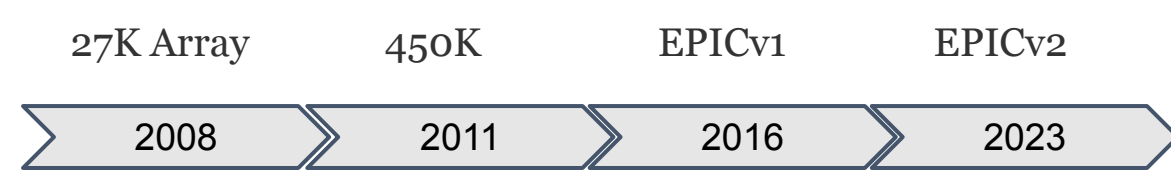


Bisulfite conversion

- Used for both array and sequencing
- $C \rightarrow U \rightarrow (\text{PCR}) \rightarrow T$
- $mC \rightarrow C \rightarrow (\text{PCR}) \rightarrow C$



DNA Methylation Array : Human Methylation BeadChip



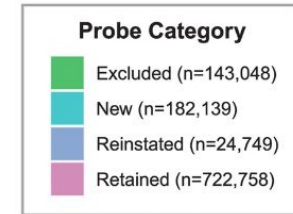
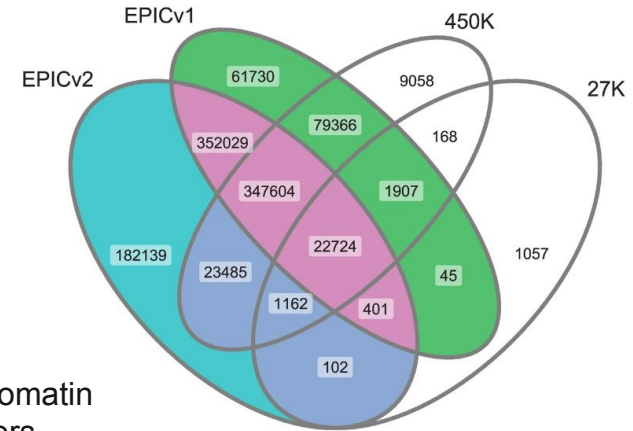
25K probes
CGI in
promoters and
cancer genes

485K probes
+
CGI shores & shelves
99% RefSeq genes
FANTOM4 promoters
MHC region
enhancers

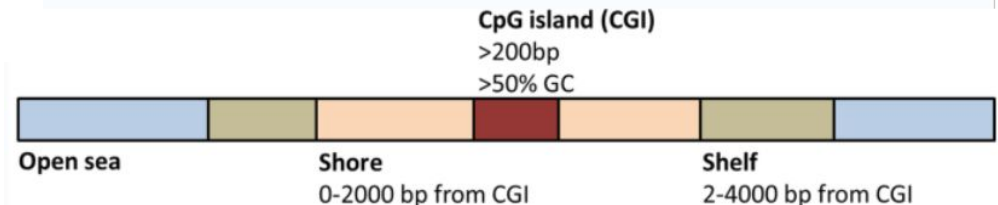
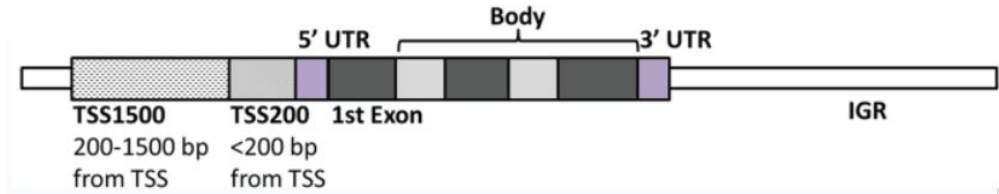
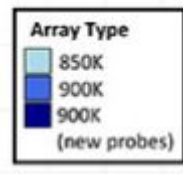
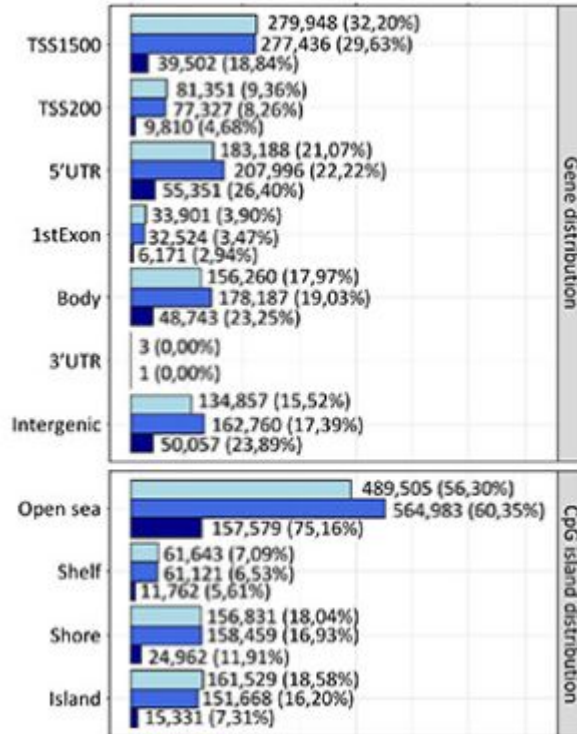
867K probes
+
FANTOM5 enhancers
miRNA promoters

935K probes
+
ENCODE open chromatin
FANTOM5 enhancers
Superenhancers
Common cancer driver
mutations

77.63% homologous to
v1
200K new probes
>99% correlation in
methylation values with
v1



DNA Methylation Array : Human Methylation BeadChip



DNA Methylation Array : Human Methylation BeadChip

- ✓ genome-wide single nucleotide resolution
- ✓ low cost, ease of use
- ✓ widely used in population scale studies
- ✓ high reproducibility and reliability
- ✓ compatible with FFPE tissues
- ✓ a few non CpG probes
- ✓ well established bioinformatics solutions

	HM450	EPICv1	EPICv2 (unique prefixes)	EPICv2 (all probes)
Total Probes	486,427	866,553	931,293	937,690
cg probes	482,421 (99.18%)	862,927 (99.6%)	926,858 (99.5%)	933,252 (99.5%)
ch probes	3,091 (0.641%)	2,932 (0.34%)	2,914 (0.31%)	2,914 (0.31%)
rs probes	65 (0.013%)	59 (0.0068%)	62 (0.0067%)	65 (0.0069%)
ct probes	850 (0.175%)	635 (0.073%)	635 (0.068%)	635 (0.068%)
nv probes	0 (0%)	0 (0%)	824 (0.88%)	824 (.088%)
Infinium-I	135,501 (27.9%)	142,158 (16.4%)	127,028 (13.6%)	128,295 (13.7%)
Infinium-II	350,926 (72.1%)	724,395 (83.6%)	804,362 (86.4%)	809,395 (86.3%)

“cg”: CpG cytosine methylation probes; “ch”: non-CG cytosine methylation probes; “rs”: common SNP probes; “nv”: probes for somatic mutations found in cancer; and “ct”: quality control probes.

Infinium Bead Technology: Type I and Type II probes

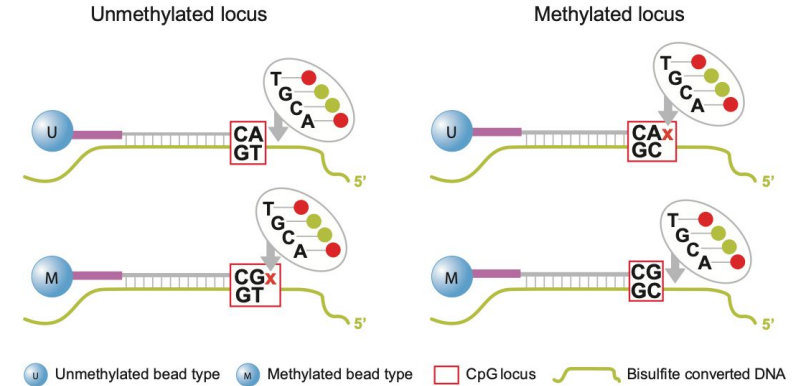
- Silica bead affixed with oligonucleotide containing 50 base probe sequence

CTACAAATACGACACCCGCAACCCATATTTTCATATATTATCTCATTTAAT

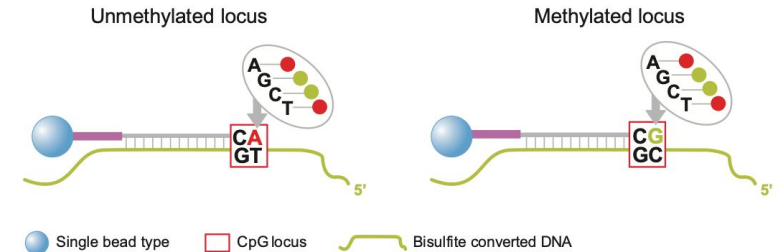
- Bisulfite converted - DNA is hybridised to the probe
- Single base extension of the probe with fluorescence ddNTP
- Fluorescence signal detects the C as thymine (T) if originally unmethylated or C if methylated.
- Infinium-I chemistry has one color channel & two bead types
 - Methylated (M) and Unmethylated (U) beads
- Infinium-II uses one bead type and two color channels

green for M, red for U

Infinium I



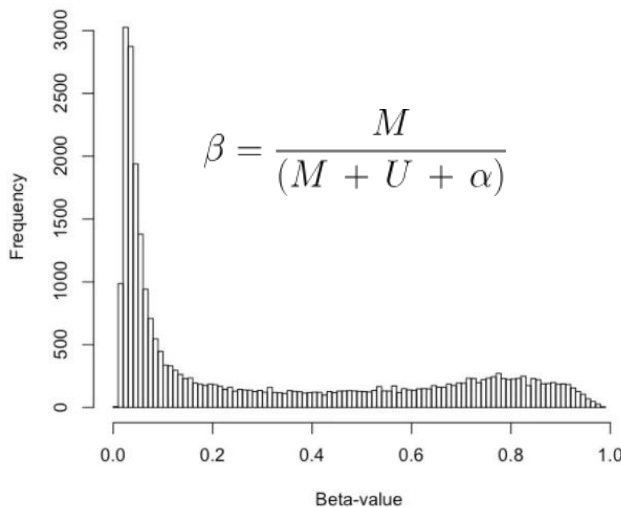
Infinium II



Measurement of Methylation Values

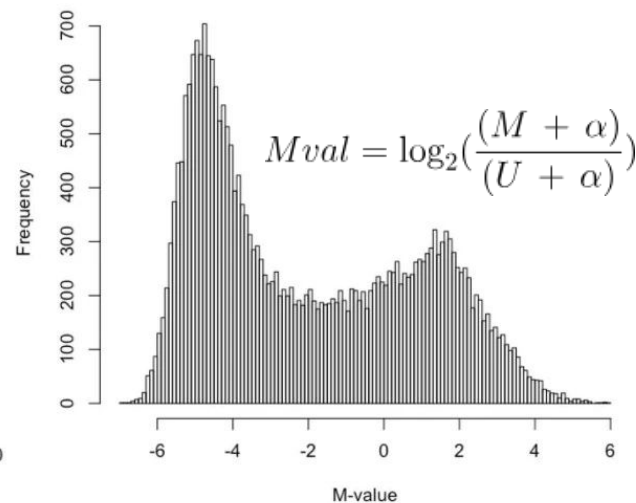
Beta value

- Bounded value between 0 to 1
- Fraction of cells with a methylated C
- Easier to interpret



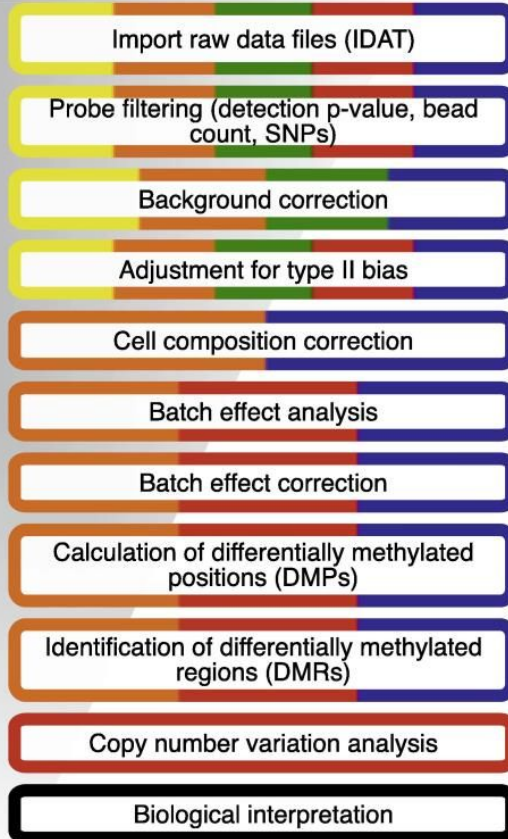
M value

- Log ratio between M and U intensities
- +M value, $M > U$
- - M value, $M < U$
- Used in statistical model testing



Analysis Pipeline

450k analysis pipeline



Freely available packages for Infinium 450k data analysis.

as of 2018

Package	Use
<i>ChAMP</i>	Comprehensive suite of functions; automated pipeline
<i>COHCAP</i>	CpG island analysis and gene expression data integration
<i>Comb-p</i>	DMR calling
<i>DMRcate</i>	DMR calling
Epigenetic clock	Predictor of sample age
<i>EWasher</i>	Reference-free cell composition correction
<i>FastDMA</i>	Quantile normalisation and DMP/DMR calling
IMA	Preprocessing including normalisation methods; Pipeline option
<i>Lumi</i>	Background correction, general normalisation
<i>Marmal-aid</i>	450k database for data integration
MethylAid	Interface for interactive sample QC
<i>Methylumi</i>	Comprehensive suite of functions
<i>Minfi</i>	Comprehensive suite of functions
<i>NIMBL</i>	Matlab code for QC and DMP calling
<i>RefFreeEWAS</i>	Reference-free cell composition correction
<i>RnBeads</i>	Comprehensive suite of functions
<i>shinyMethyl</i>	Interface for interactive sample QC
<i>watermelon</i>	Preprocessing including performance metrics and numerous normalisation methods

[SeSAMe](#)

Import raw methylation data

- Raw IDAT files are in folder named after chip ID
- Red/Green signal intensity files per sample

- IDAT file name format

5975827018_R06C02_Grn.idat

5975827018_R06C02_Red.idat

<chip barcode>_<chip position>_<channel>

- Sample annotation CSV file

```
1 dataDirectory <- "/sw/courses/epigenomics/DNAMethylation/array_data/"
2 # read in the sample sheet for the experiment
3 targets <- read.metharray.sheet(dataDirectory, pattern="SampleSheet.csv")
4 # read in the raw data from the IDAT files
5 rgSet <- read.metharray.exp(targets=targets)
6 # Go from intensity data to methylation levels
7 MSet <- preprocessRaw(rgSet)
```

Quality Control to flag poor quality and outlier samples

Median intensity of M vs U

Beta value distribution

Probes detection P value

Internal control probes

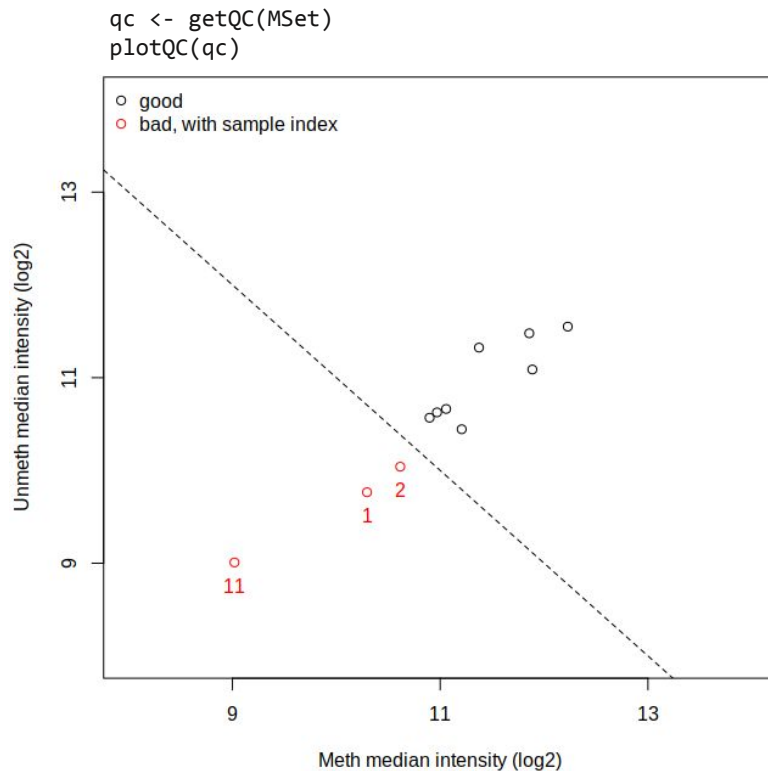
Bisulfite conversion

Hybridization

Extension

Negative controls

Gender check



Quality Control to flag poor quality and outlier samples

Median intensity of M vs U

Beta value distribution

Probes detection P value

Internal control probes

Bisulfite conversion

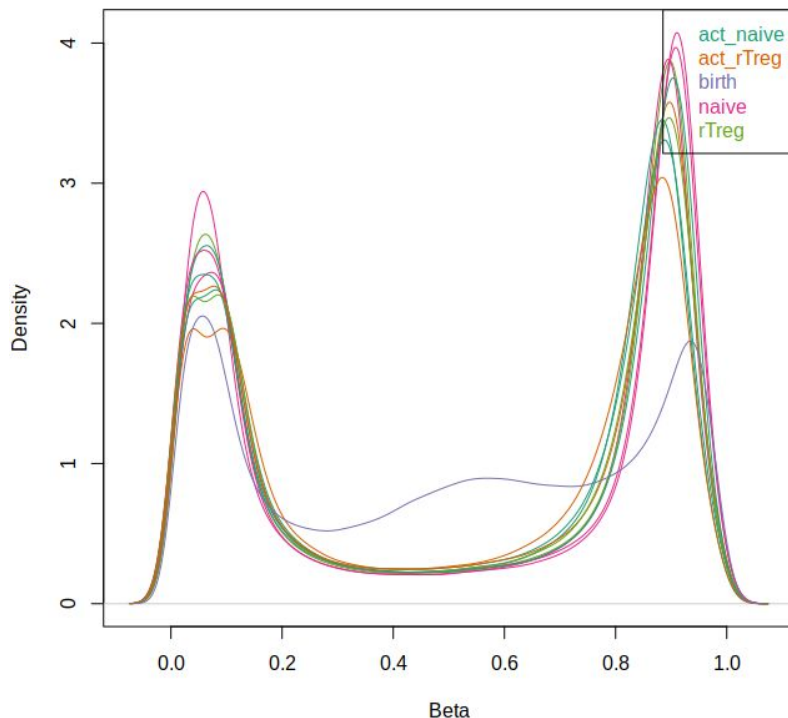
Hybridization

Extension

Negative controls

Gender check

```
phenoData <- pData(MSet)
densityPlot(MSet, sampGroups = phenoData$Sample_Group)
```



Quality Control to flag poor quality and outlier samples

Median intensity of M vs U

Beta value distribution

Probes detection P value

Internal control probes

Bisulfite conversion

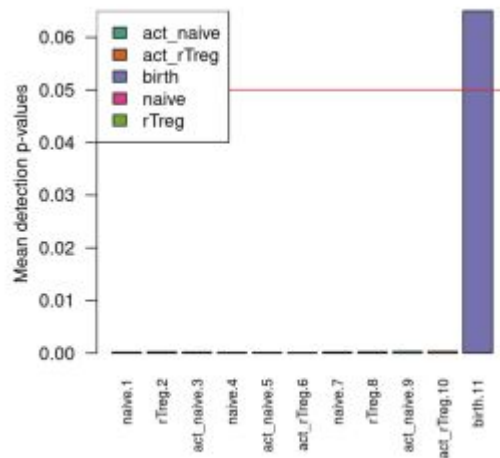
Hybridization

Extension

Negative controls

Gender check

```
1 # Calculate the detection p-values
2 detP <- detectionP(rgSet)
3 # examine mean detection p-values across all samples to identify any failed
4 barplot(colMeans(detP), las=2, cex.names=0.8, ylab="Mean detection p-values")
5 abline(h=0.05,col="red")
```



Quality Control to flag poor quality and outlier samples

Median intensity of M vs U

Beta value distribution

Probes detection P value

Internal control probes

Bisulfite conversion

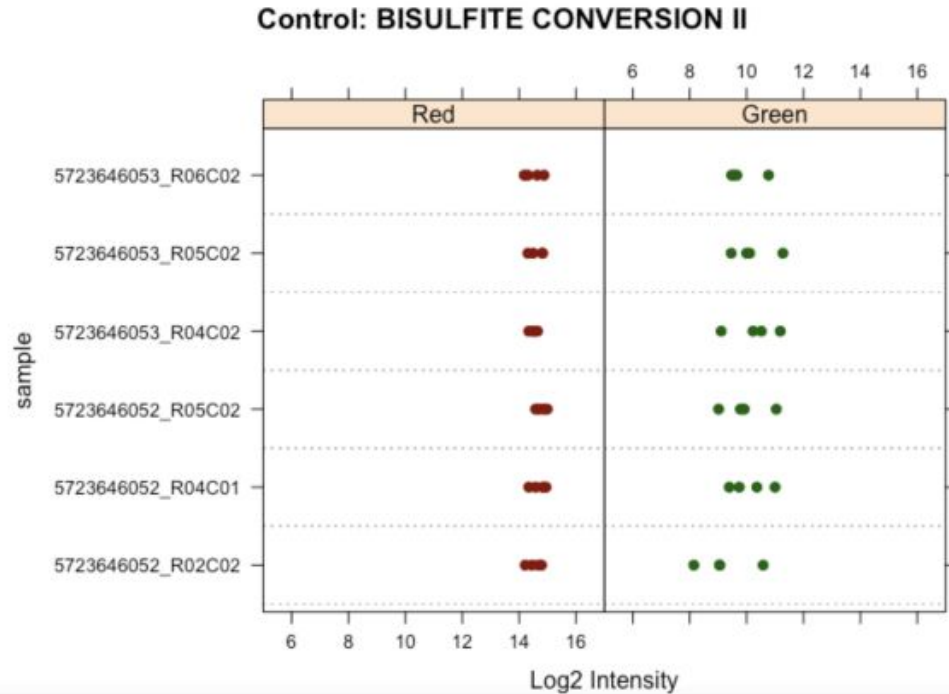
Hybridization

Extension

Negative controls

Gender check

```
1 controlStripPlot(RGSet, controls="BISULFITE CONVERSION II")
```



Quality Control to flag poor quality and outlier samples

Median intensity of M vs U

Beta value distribution

Probes detection P value

Internal control probes

Bisulfite conversion

Hybridization

Extension

Negative controls

Gender check

QC Probes

Control Probe	Purpose
Staining	measure efficiency and sensitivity of staining step (independent of hybridisation/extension step)
Extension	test efficiency of extension of A, T, C and G nucleotides from a hairpin probe (sample-independent). The perfect match hairpin controls should result in high signal, and the mismatch probes in low signal
Hybridization	test the overall performance of Infinium assay using synthetic targets (not DNA) at 3 concentrations
Target removal	test efficiency of stripping step after extension
Bisulphite conversion	test efficiency of bisulphite conversion by query of C/T polymorphism
Specificity controls	check for non-specific detection of methylation signal over unmethylated background. Specificity controls are designed against non-polymorphic T sites (G/T mismatch)
Non-polymorphic	query a non polymorphic base A, T, C and G to test overall performance of the assay from amplification to detection
Negative	randomly permuted bisulphite-converted sequences containing no CpGs. They should not hybridise to DNA. The mean of these probes determines the system background

Quality Control to flag mislabelled samples

Median intensity of M vs U

Beta value distribution

Probes detection P value

Internal control probes

Bisulfite conversion

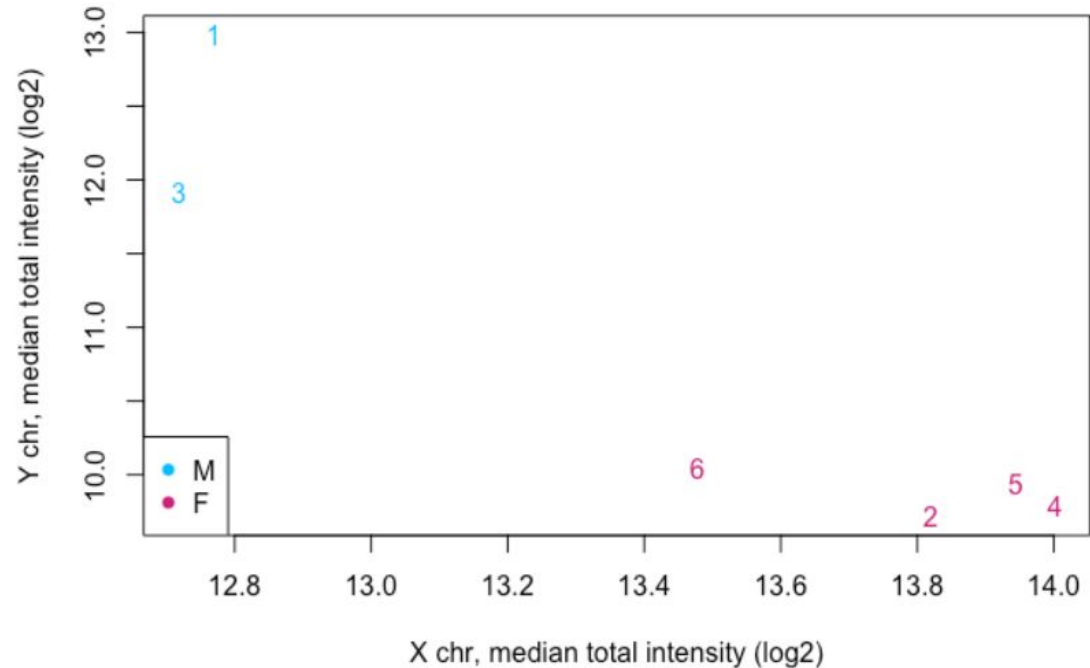
Hybridization

Extension

Negative controls

Gender check

```
predictedSex <- getSex(GRset, cutoff = -2)$predictedSex  
plotSex(getSex(GRset, cutoff = -2))
```



Probe Filtering

remove probes with high detection P value

median $P > 0.01$ across samples, $P > 0.01$ in nth% of samples

remove probes overlapping SNPs

`minfi::dropLociWithSnps`

$MAF > 0.05$

drop probes in X, Y chromosome

remove cross reactive probes

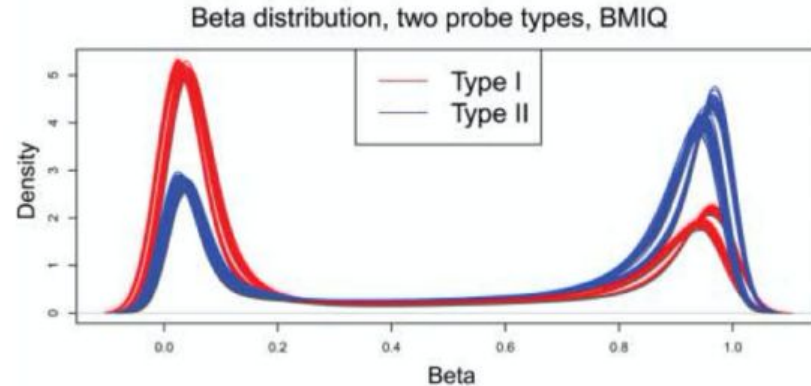
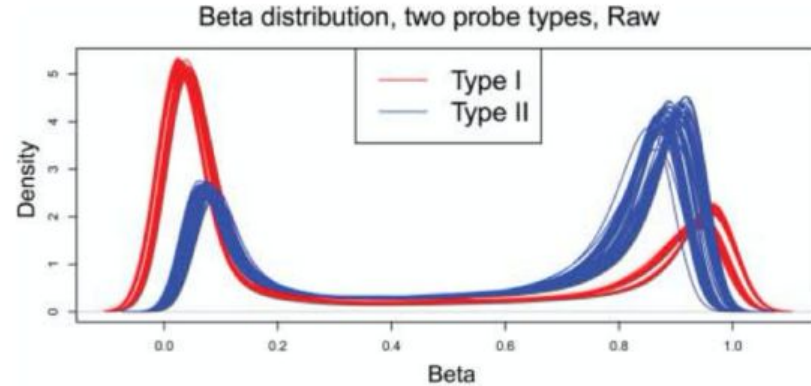
Preprocessing, correction and normalisation

Within array:

- Dye bias correction
- Background correction
- Type I/II probe bias correction

Across array:

- starting material
- labelling efficiency

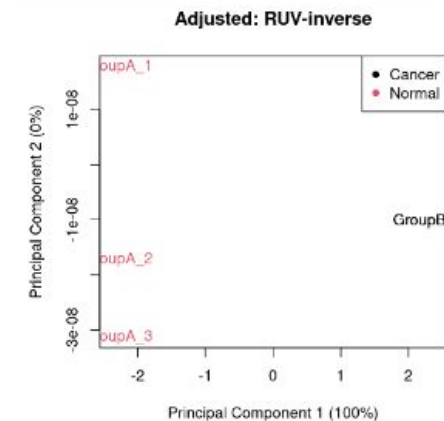
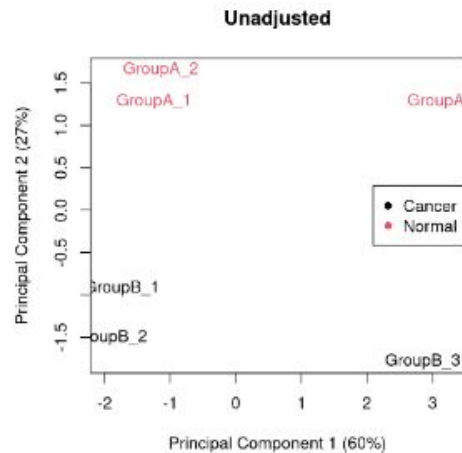


Normalisation in minfi

	Dye Bias	Background Correction	Type I/II Probe bias	Within array normalisation	Across array normalisation
preprocessRaw	-	-	-	-	-
preprocessIllumina	✓	✓	-	✓	✓
preprocessSWAN	-	-	✓	✓	-
preprocessQuantile	-	-	✓	✓	✓
preprocessNoob	✓	✓	-	-	-
preprocessFunnorm	✓	✓	✓	✓	✓

Differentially Methylated Probes (DMPs)

- Perform a statistical test to find any significant association between the methylation state of a CpG and the phenotype of interest
 - T-test, ANOVA
 - Wilcoxon rank-sum, Kruskal Wallis test, Fisher's exact test
 - Linear model, logistic regression, mixed effects model, Beta Binomial
- `minfi` uses `limma` functionality
 - Use M-values
- Phenotype could be categorical or continuous
 - cancer vs normal, between tissue types, smokers vs not
 - age, blood pressure, BMI
- Other sources of variation can be accounted for
 - E.g., plate-to-plate, lot-to-lot variance
 - covariate in the model
 - ComBat
 - SVA (Surrogate Variable Analysis)
 - RUVSeq



Differentially Methylated Regions (DMRs)

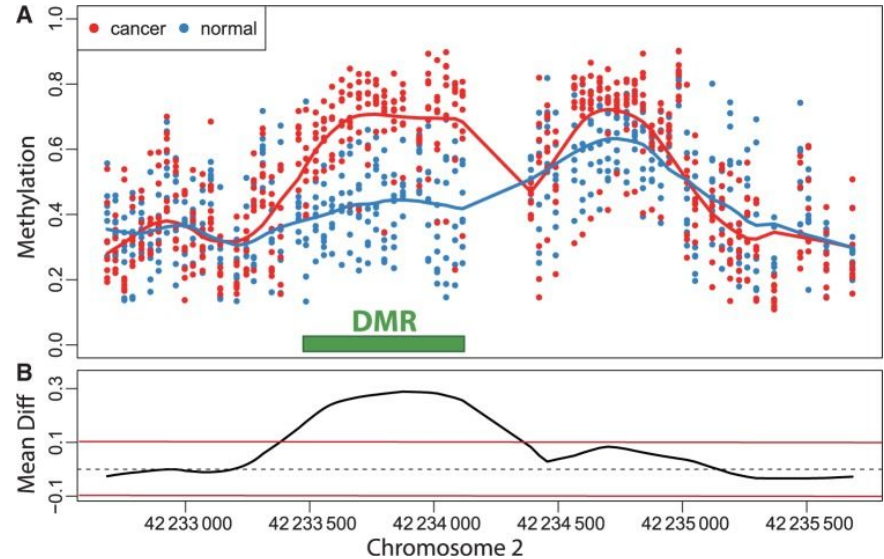
DMRs can be defined:

heuristically / De novo

[Bumphunter](#), [DMRcate](#),
[ProbeLasso](#), [comb-p](#)

functional units

CpG islands, gene bodies,
promoters



WGBS

Sequencing-based methylation profiling

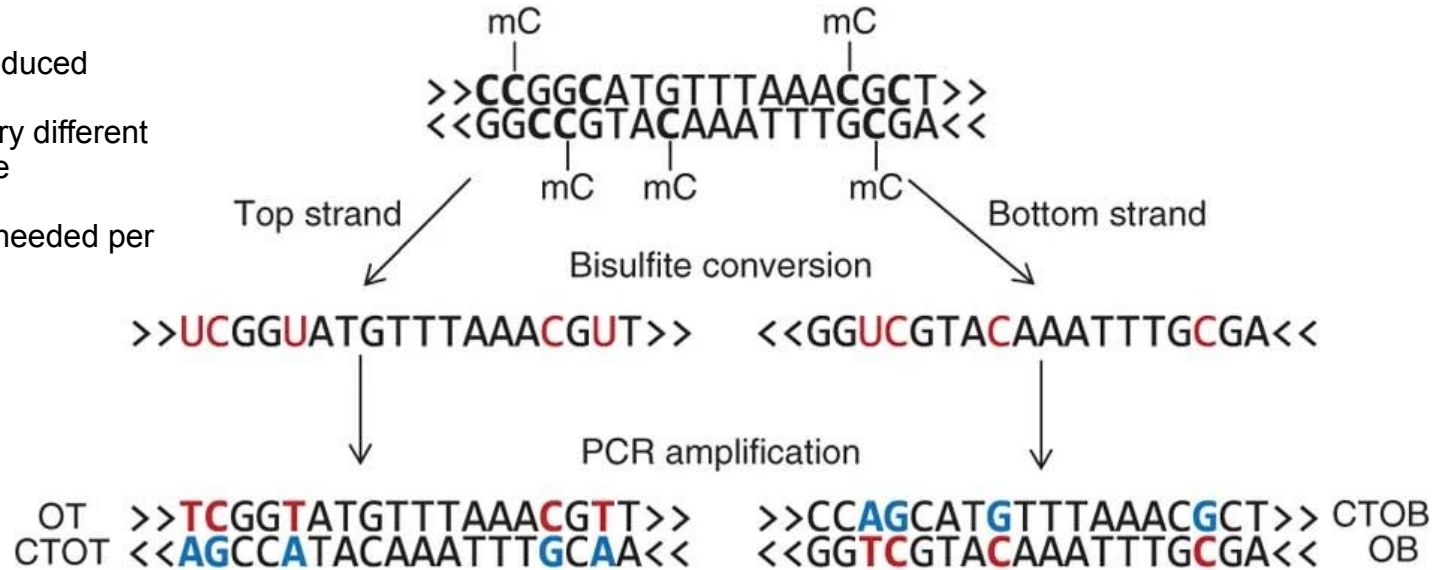
	Enzyme digestion	Affinity enrichment	Sodium bisulfite
Principles	Some restriction enzymes, such as <i>HpaII</i> and <i>SmaI</i> , are inhibited by 5 ^{me} C in the CpG.	Affinity enrichment uses antibodies specific for 5 ^{me} C or methyl-binding proteins with affinity for profiling of DNA methylation.	Sodium bisulfite chemically turns unmethylated cytosine into uracil, hence enabling methylation detection.
Method example	Methyl-seq *MCA-seq *HELP-seq *MSCC	*MeDIP-seq *MIRA-seq	*RRBS *WGBS *BSPP

*MCA: methylated CpG island amplification; *HELP: HpaII tiny fragment enrichment by ligation-mediated PCR; *MSCC: methylation-sensitive cut counting; *MeDIP-seq: methylated DNA immunoprecipitation; *MIRA: methylated CpG island recovery assay; *RRBS: reduced representation bisulfite sequencing; *WGBS: whole genome bisulfite sequencing; *BSPP: bisulfite padlock probes.

Read mapping of bisulfite treated DNA

Challenges:

- BS treatment renders reduced genome complexity
- BS treated reads are very different from the reference sequence
- Up to 4 read alignment needed per locus



2 different PCR products and 4 different sequence strands from one genomic locus
 Each of these 4 sequence strands can theoretically exist in any possible conversion state

3 letter alignment of BS-converted reads

sequence of interest

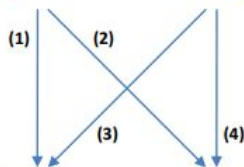
TTGGCATGTTTAAACGTT

C>T

G>A

5' ...TTGGTATGTTTAAATGTT...3'

5' ...TTAACATATTTAAACATT...3'



...TTGGTATGTTTAAATGTT...

...AACCATACAAATTTACAA...

forward strand C -> T converted genome

...CCAACATATTTAAACACT...

...GGTTGTATAAATTTGTGA...

forward strand G -> A converted genome
(equals reverse strand C -> T conversion)

(1) (2) (3) (4)

5' ...CCGGCATGTTTAAACGCT...3'

read sequence TTGGCATGTTTAAACGTTA

genomic sequence CCGGCATGTTTAAACGCTA

methylation call xZ . . H Z . h . .

Fully bisulfite convert read
(as both forward and reverse strand)

Align to bisulfite converted genomes

Read all 4 alignment outputs and extract
the unmodified genomic sequence if the
sequence could be mapped uniquely

Methylation Call

h unmethylated C in CHH context

H methylated C in CHH context

x unmethylated C in CHG context

X methylated C in CHG context

z unmethylated C in CpG context

Z methylated C in CpG context



Bismark

**after alignment*

chromosome

position

HISEQ2000-06:366:C3G4NACXX:3:1101:1316:2067_1:N:0:	99	16	71322125	255	100M	=
71322232	-207					
NTTATTTTAGTTTTTtagggTTtGTGTgTAGGAGTGtGGGAATTATGTTTTTTATGGTTGATATTTATTTAAAAGTGAGTATAAATATATATATTTTTTT ← sequence						
#1=DDDDDAAFFHIIIA:<FGHCCEFGHD?CFFBBBGEHHGHIII<FEHIIIII==DE??EHHFHEEEEEEEEC>;>66:@CDEEDDCEEEEEEDDDCB						quality
NM:i:14 XX:Z:G8C2C7C21C13C6CC1C17CC3C4CC4						
XM:Z:.....h..h.....x.....h.....x.....hh.h.....hh...h...hh...						
XR:Z:CT XG:Z:CT XA:Z:l						
HISEQ2000-06:366:C3G4NACXX:3:1101:1316:2067_1:N:0:	147	16	71322232	255	100M	=
71322125	-207					
GGTTATTTTATTTAGGGTTATTGTTTTAGAGTTTTATTGTTGTGAACAGATATATGATTAAGGTAATTTTATAAGGATAATATTTAATTGGAGTTGGTT						
CCCCEECADCFFFFFHGHGHGHIIGIHFIIJJIJHFGHGGEHIIJIGIGFJJJJJJJJJJIGJJJGJJJJIIIJJIJJJJJJJHNNHHHFFFFFFCC						
NM:i:21 XX:Z:2G2CC1C1C1C1C1C2C10C1C4CC4C2C1C3C5C2C1C2C3C1						
XM:Z:....hh.h.h.x.....h.....x..x.....X...h.h...hh...h..h.h...h...h...x...h.						
XR:Z:GA XG:Z:CT XB:Z:l						

methylation
call

Read 2

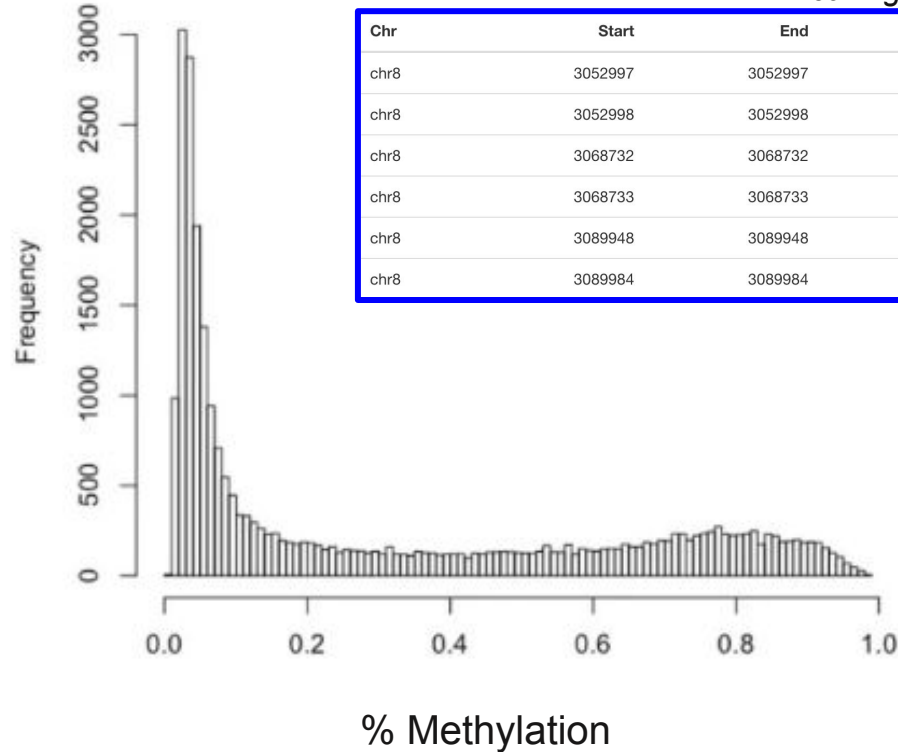
methylation call

5' - **TT**GG^{me}CATGTTTAAAC^{me}GT**T** - 3' bisulfite read
 5' ...**cc**gg**c**atgttttaa**a**cg**c**t...3' genomic sequence
 ↓ ↓ ↓ ↓ ↓
 xz...**H**.....**Z**.h. methylation call

z	unmethylated	C	in CpG context
Z	methylated	C	in CpG context
x	unmethylated	C	in CHG context
X	methylated	C	in CHG context
h	unmethylated	C	in CHH context
H	methylated	C	in CHH context

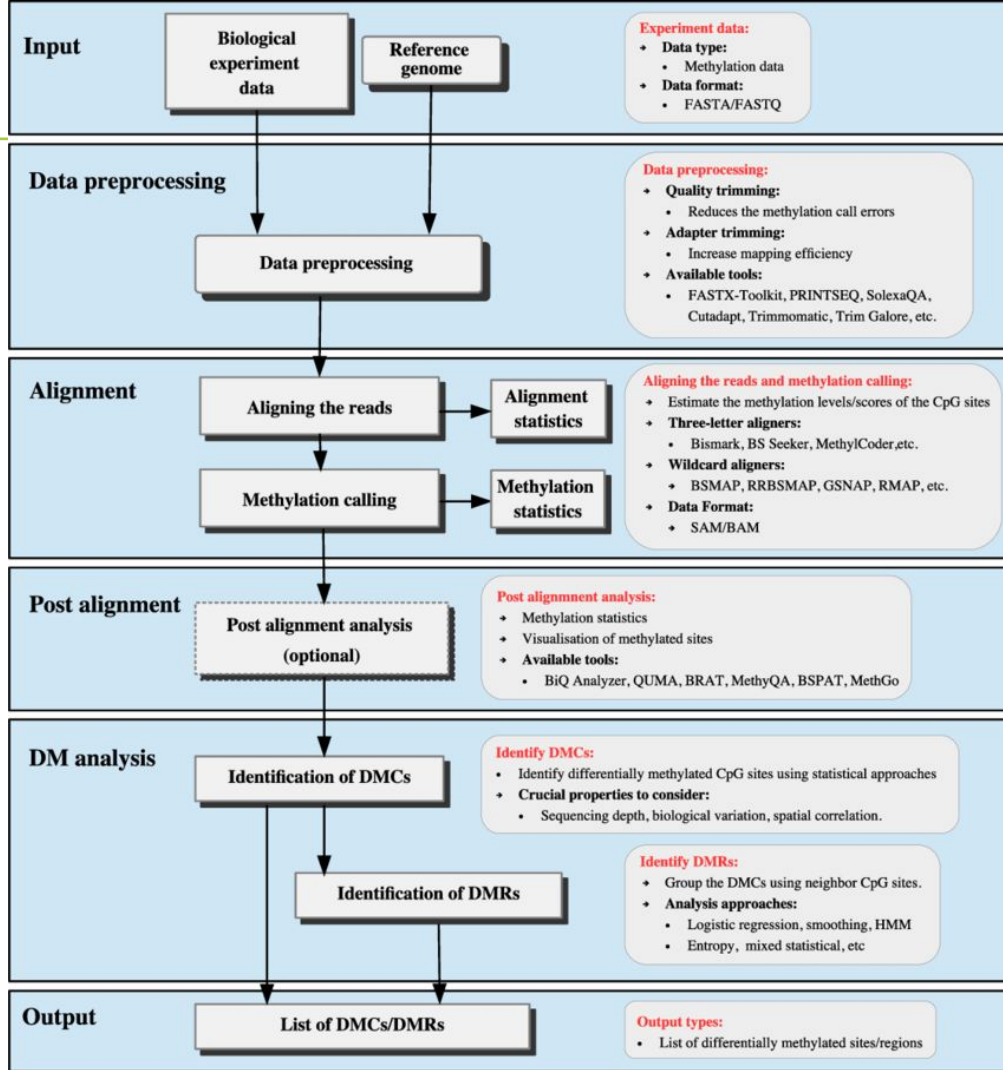
BS-seq output* is %Methylation of a cytosine

**after alignment and methylation calling*

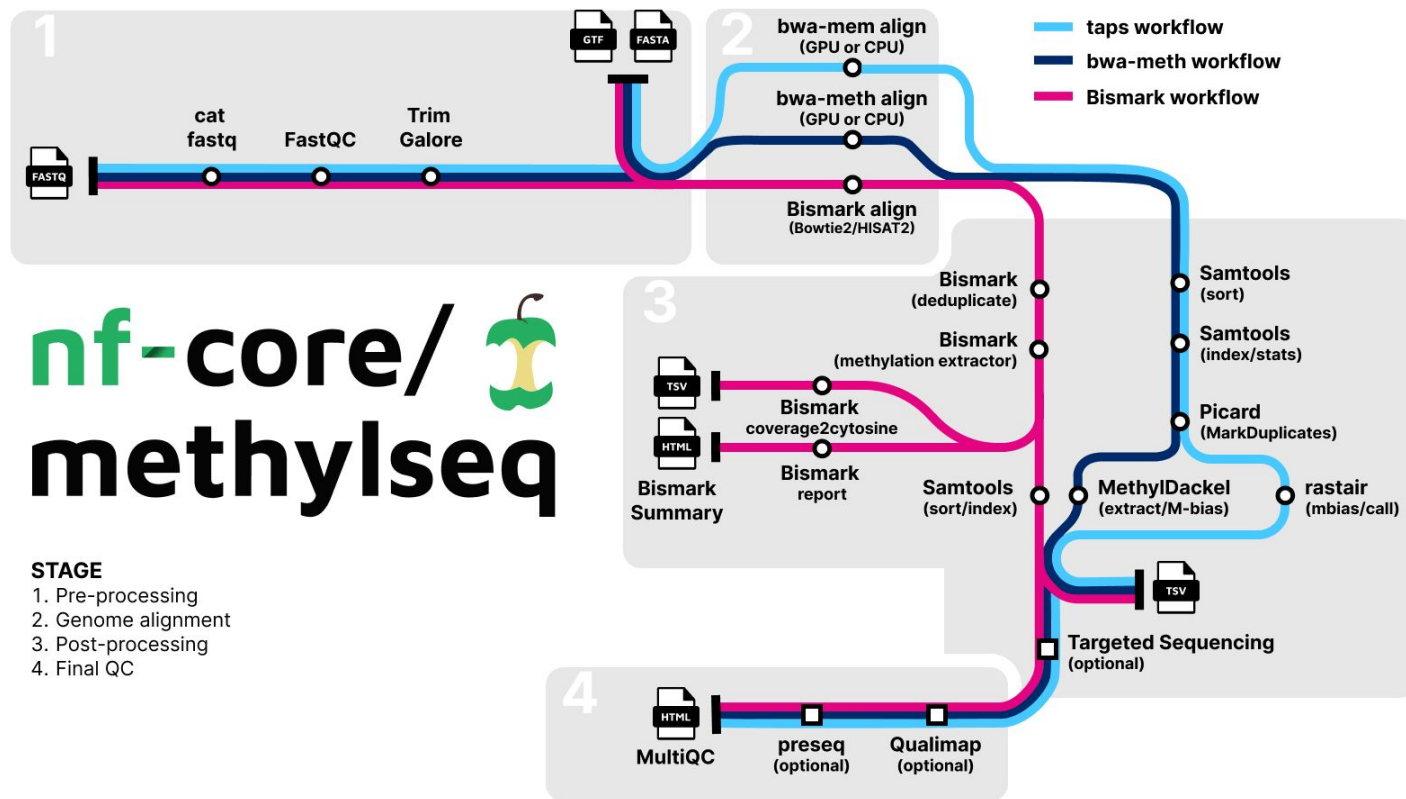


Analysis pipeline

nf-core/methylseq



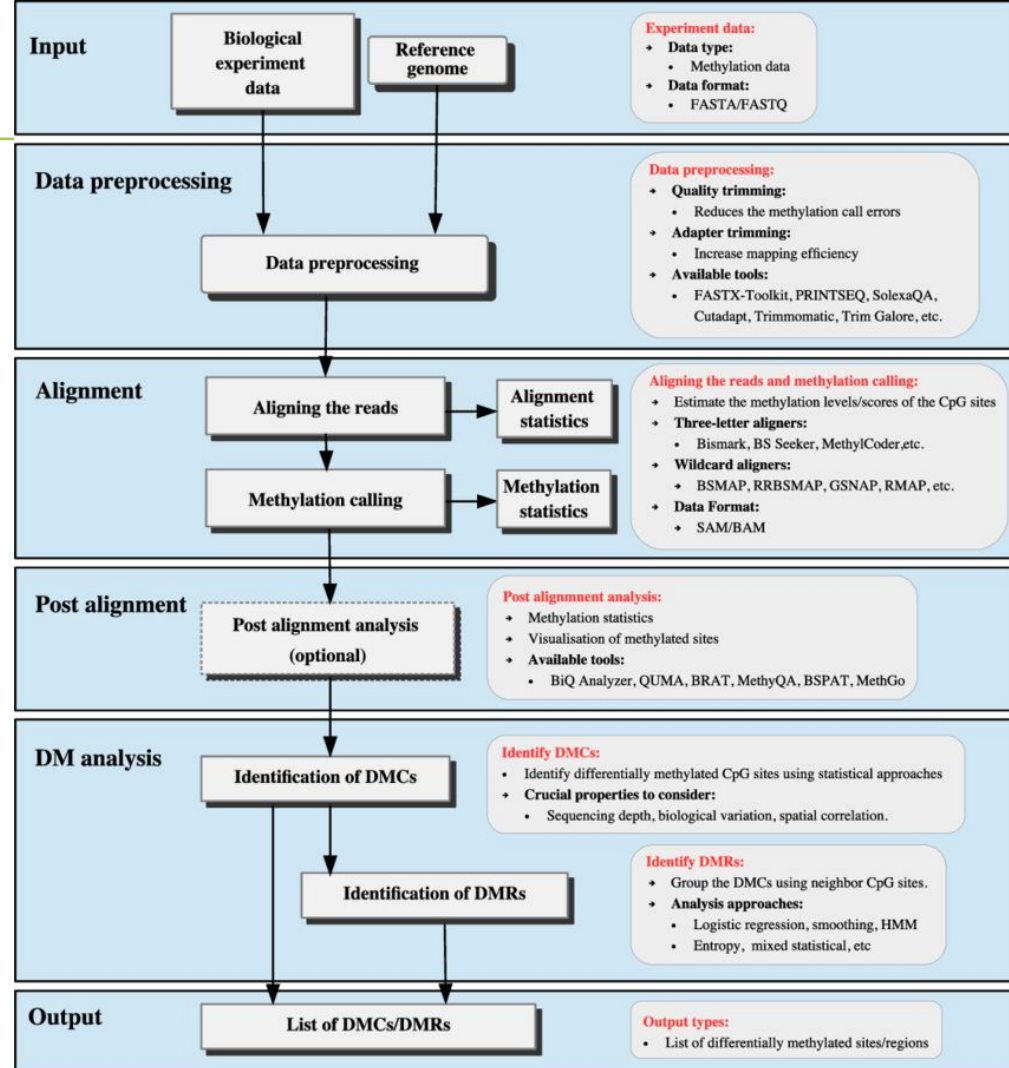
nf-core/methylseq is a bioinformatics analysis pipeline for BS-seq data



Analysis pipeline

nf-core/methylseq

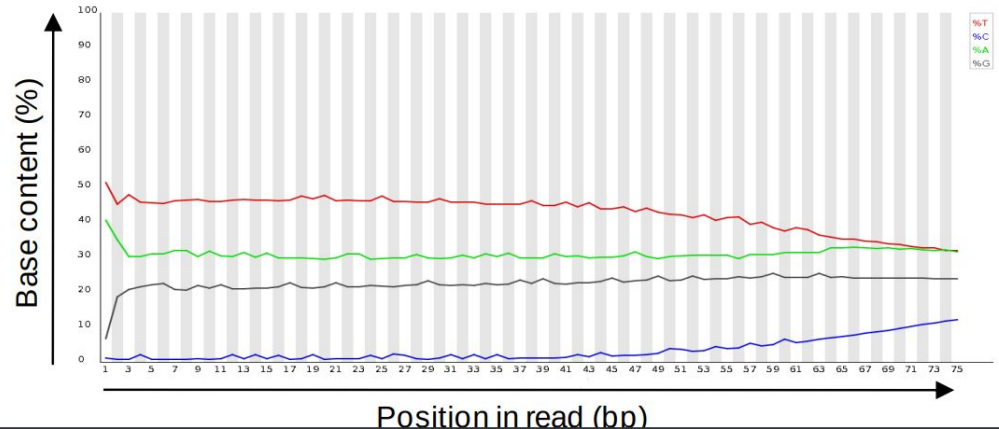
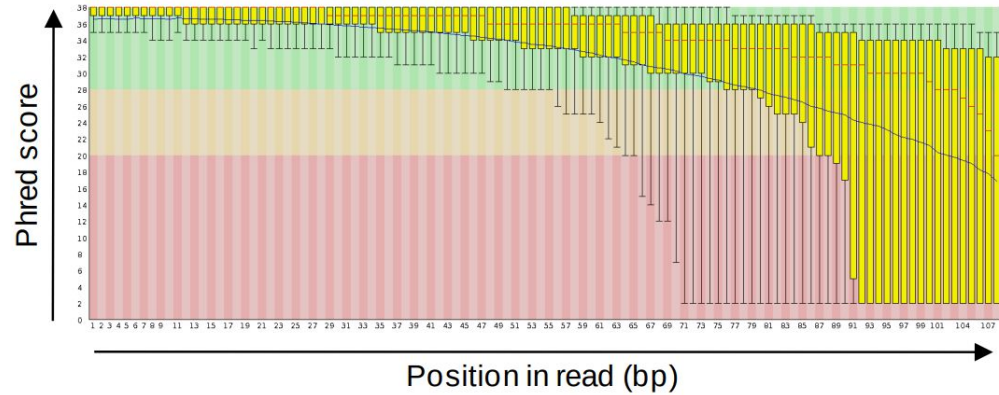
Exercise for today



Quality control: base call quality

Trim reads of low quality base calls and adapter sequence

- to increase mapping
- to avoid false M calls



5' -
3' -

-3'
-5'

G**G****G**NNNNNNNNNNNNNNNNNNNNNNNNNNNNNN**C****C****A**
A**C****C****C**NNNNNNNNNNNNNNNNNNNNNNNNNNNNNN**G****G**



mapped Quality control

Bisulfite conversion efficiency

- non CpG sites in mammalian genome* should have > 99.5% conversion in a good experiment
- Spike-in controls e.g., phage Lambda

DNA degradation during bisulfite conversion

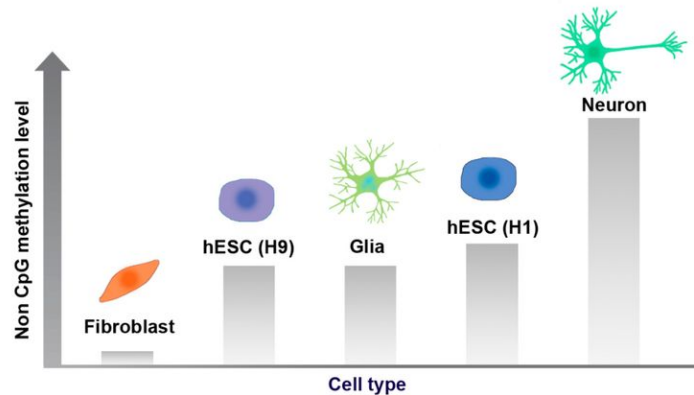
- unique alignment rates
- read length after trimming

Remove C>T SNPs

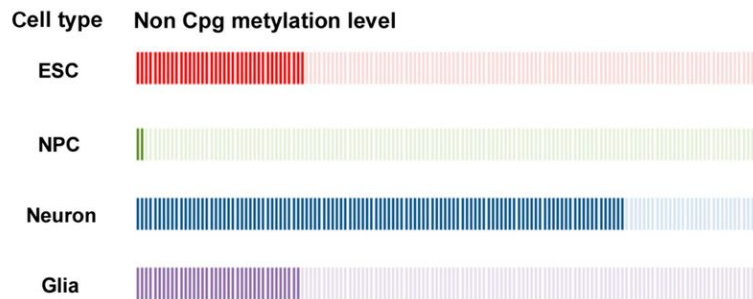
Deduplication

- recommended for WGBS, not for RRBS

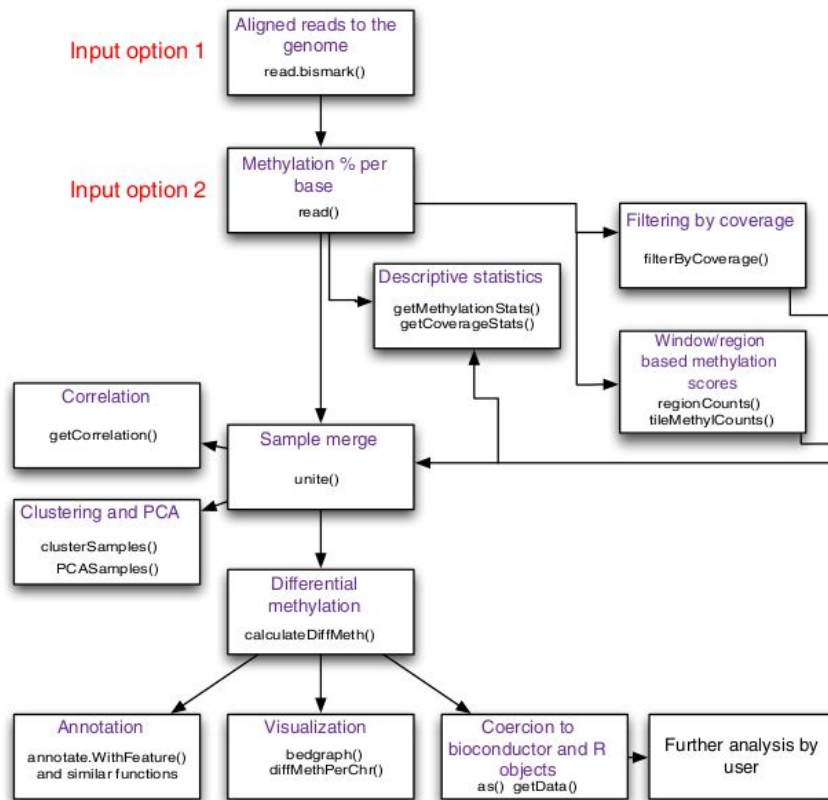
A



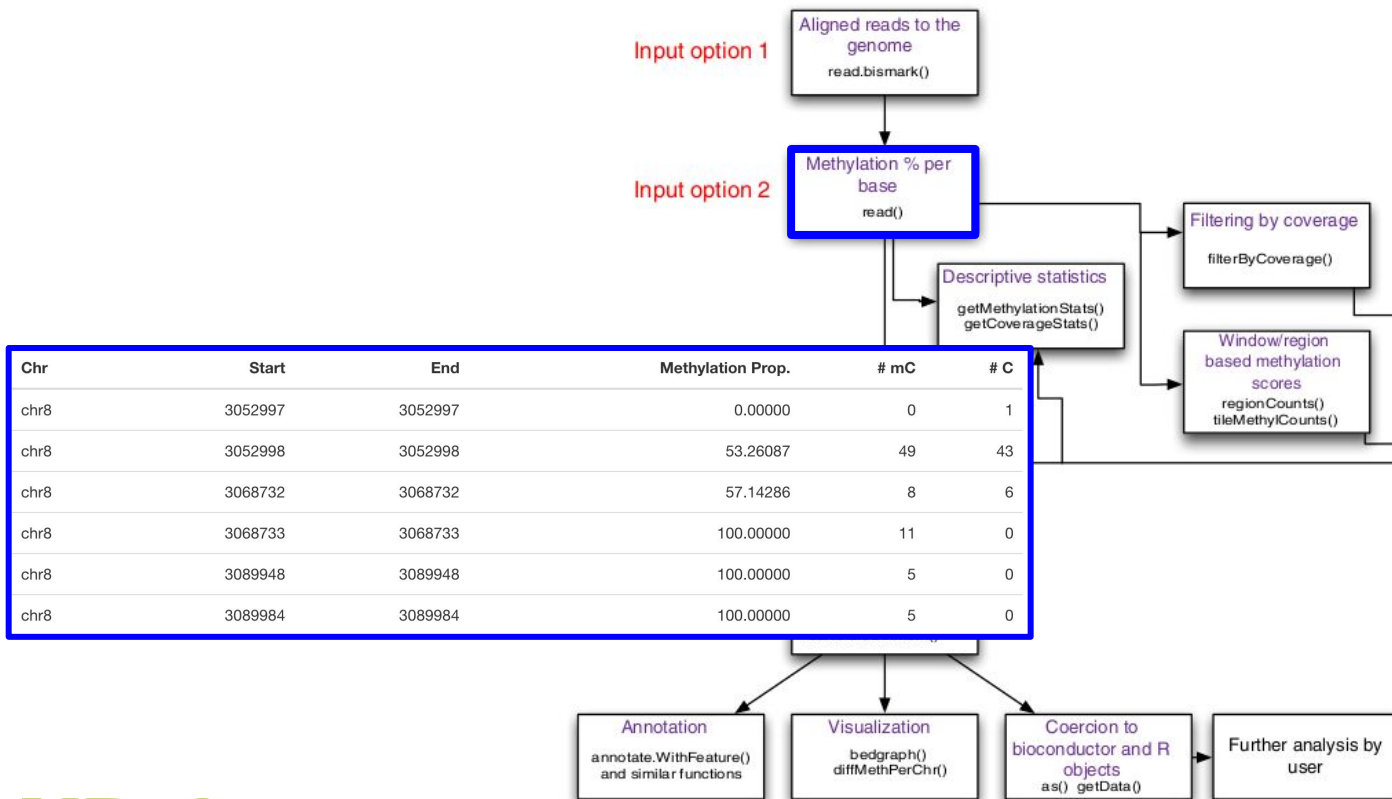
B



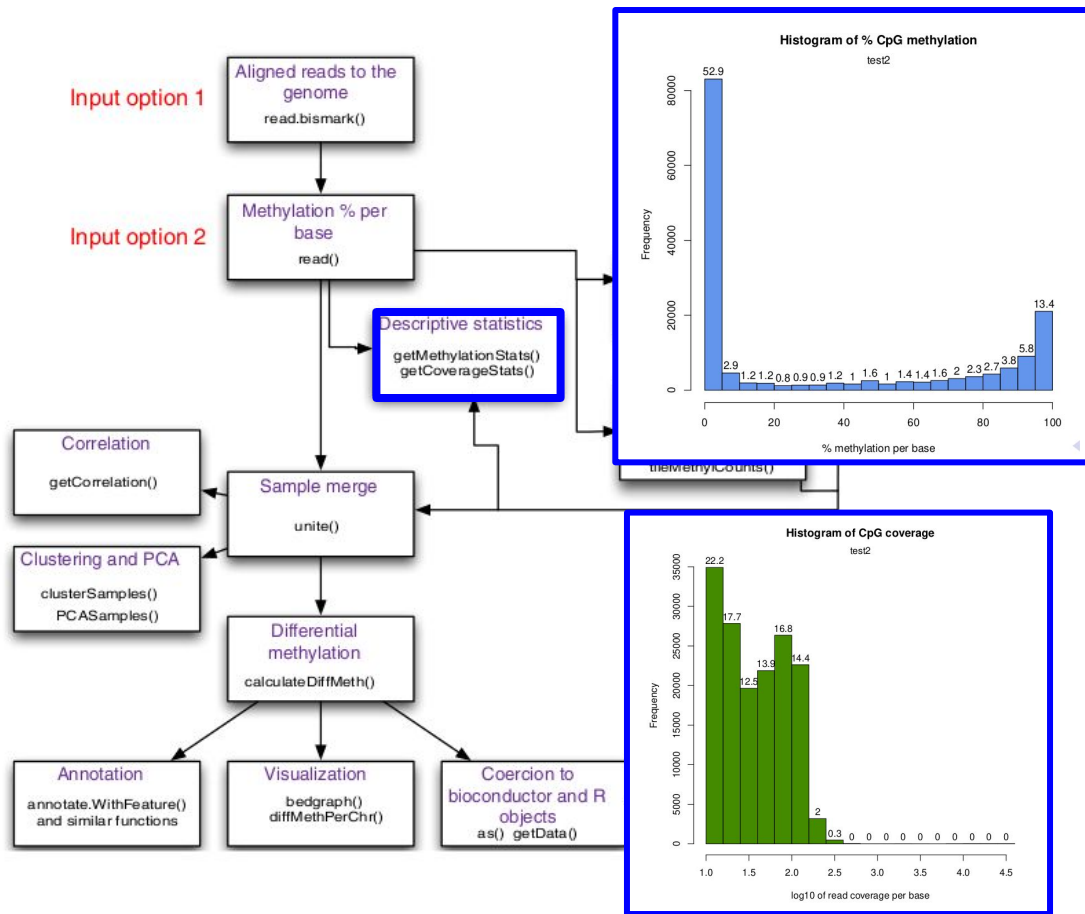
Analysis workflow for WGBS data



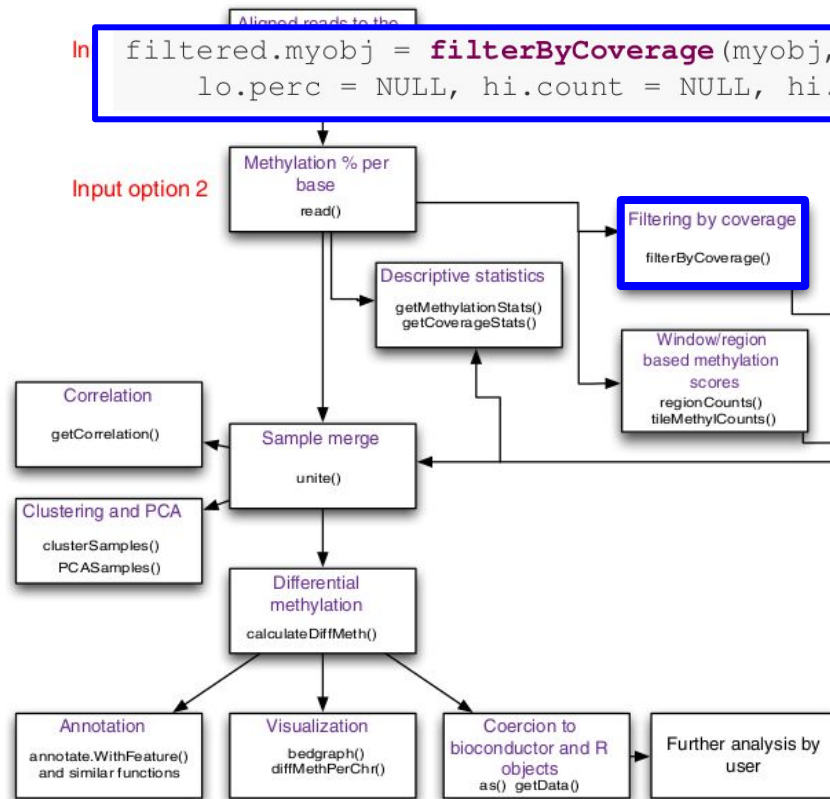
Analysis workflow for WGBS data



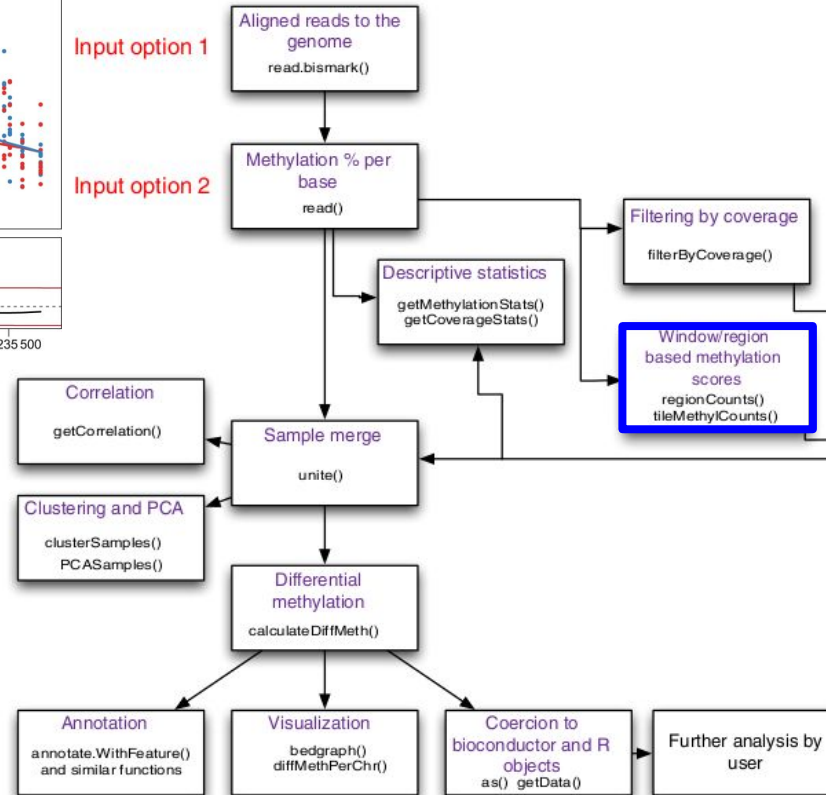
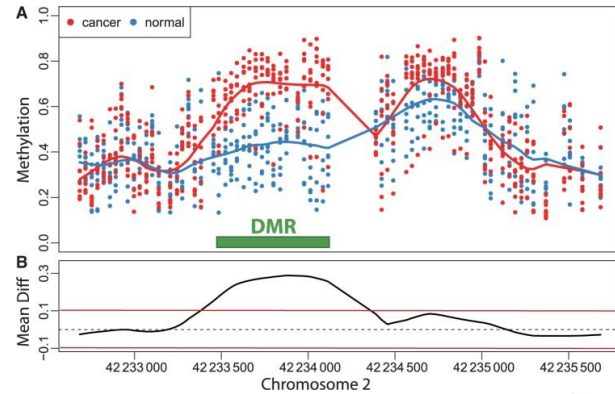
Analysis workflow for WGBS data



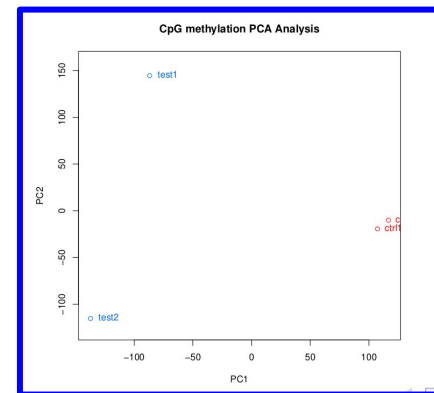
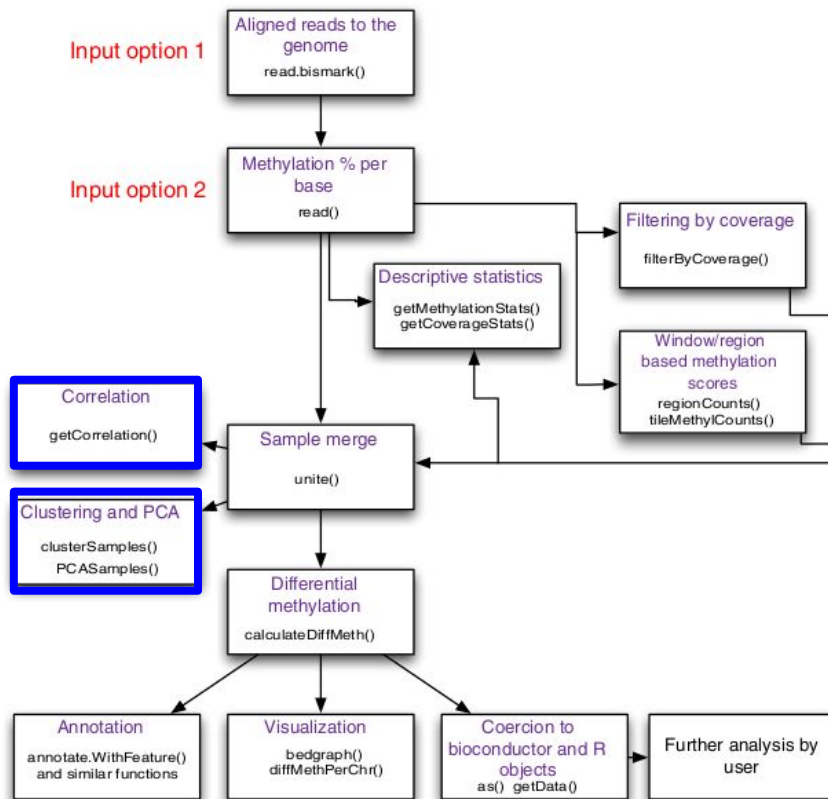
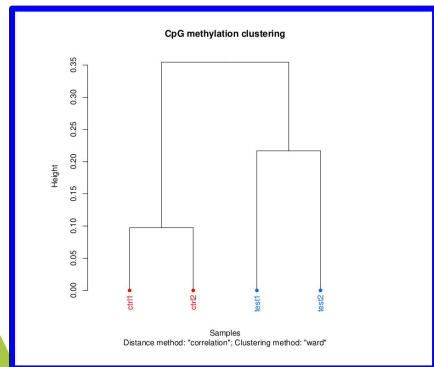
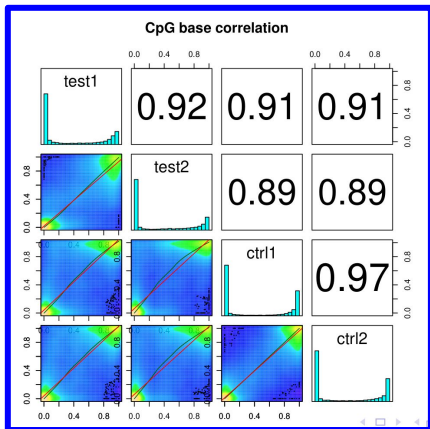
Analysis workflow for WGBS data



Analysis workflow for WGBS data



Analysis workflow for WGBS data



Differential Methylation

Remove CpGs with little variation

Remove CpGs that overlap C>T SNPs

No replicates: Fisher's exact test

With replicates:

Logistic regression

Beta Binomial

Overdispersion correction

Covariates can be included in the model

